cURL Sikho — A to Z Guide

Hinglish mein, practical examples ke saath

Reference API: `https://jsonplaceholder.typicode.com/todos/1`

Is poore guide mein GET, POST, PUT, DELETE, headers, verbose mode, aur output save karna — sab kuch isi API ko use karke samjhaya gaya hai.

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1. curl kya hota hai?

curl ek command-line tool hai jiske through hum servers ko HTTP (ya FTP, etc.) requests bhej sakte hain — bina browser khole. Yeh developers, testers aur DevOps engineers ke liye sabse zyada use hone wala tool hai APIs test karne ke liye.

Basic syntax hamesha yeh hota hai:

```
curl [options] [URL]
```

Matlab — **curl** command likho, uske baad kuch **options** (jaise -X, -H, -d) aur last mein woh **URL** jahan request bhejni hai.

2. Sabse Simple GET Request

Kisi bhi API se data mangwane (fetch karne) ke liye sirf URL de do — curl by default GET request bhejta hai:

```
curl https://jsonplaceholder.typicode.com/todos/1
```

Output (server se yeh JSON response wapas aayega):

```
{
  "userId": 1,
  "id": 1,
  "title": "delectus aut autem",
  "completed": false
}
```

Yahan **todos/1** ka matlab hai — id number 1 wala todo item fetch karo.

3. Response ko Sundar (Pretty) Dikhana

Bade JSON responses ek line mein ugly dikhte hain. Isko readable banane ke liye **-s** (silent, progress bar hata do) aur pipe karke **python -m json.tool** ya **jq** use karo:

```
curl -s https://jsonplaceholder.typicode.com/todos/1 | python3 -m json.tool
```

4. Headers aur Verbose Mode Dekhna

-i — Response ke headers bhi dikhao (body ke saath):

```
curl -i https://jsonplaceholder.typicode.com/todos/1
```

Output mein sabse upar **HTTP/1.1 200 OK**, Content-Type, Content-Length jaise headers dikhenge, phir neeche JSON body.

-I (capital) — Sirf headers chahiye, body nahi:

```
curl -I https://jsonplaceholder.typicode.com/todos/1
```

-v — Verbose mode: pura connection process dikhta hai — DNS lookup, TCP handshake, request headers jo bheje gaye, response headers jo mile:

```
curl -v https://jsonplaceholder.typicode.com/todos/1
```

Debugging ke waqt (jaise: request fail ho rahi hai, ya wrong response aa raha hai) yeh sabse pehla tool hai jo use karna chahiye.

5. Custom Headers Bhejna (-H)

Agar API ko koi specific header chahiye (jaise Authorization token ya Content-Type), toh **-H** option use karo:

```
curl -H "Content-Type: application/json" \
-H "Authorization: Bearer YOUR_TOKEN" \
https://jsonplaceholder.typicode.com/todos/1
```

Multiple headers chahiye ho toh **-H** ko baar-baar likh sakte ho.

6. POST Request — Naya Data Bhejna

Naya resource create karne ke liye **-X POST** aur **-d** (data) use karte hain:

```
curl -X POST https://jsonplaceholder.typicode.com/todos \
-H "Content-Type: application/json" \
-d '{"userId": 1, "title": "curl seekhna", "completed": false}'
```

Server is tarah ka response dega (naya id assign karke):

```
{
  "userId": 1,
  "title": "curl seekhna",
  "completed": false,
  "id": 201
}
```

Note: jsonplaceholder ek fake/testing API hai — yeh data asal mein save nahi hota, bas ek simulated response deta hai. Practice ke liye perfect hai.

7. PUT aur PATCH — Data Update Karna

PUT pura resource replace karta hai, **PATCH** sirf diye gaye fields update karta hai:

```
curl -X PUT https://jsonplaceholder.typicode.com/todos/1 \
-H "Content-Type: application/json" \
-d '{"id": 1, "userId": 1, "title": "updated title", "completed": true}'
```

```
curl -X PATCH https://jsonplaceholder.typicode.com/todos/1 \
-H "Content-Type: application/json" \
-d '{"completed": true}'
```

8. DELETE — Resource Hatana

```
curl -X DELETE https://jsonplaceholder.typicode.com/todos/1
```

Success hone par server generally empty response **{}** ya status code 200/204 deta hai.

9. Response ko File Mein Save Karna

-o filename — output ko file mein save karo:

```
curl -o todo1.json https://jsonplaceholder.typicode.com/todos/1
```

-O (capital) — URL ke last part ko hi filename bana do:

```
curl -O https://jsonplaceholder.typicode.com/todos/1
```

10. Zaroori Options — Quick Reference

Option	Kaam
`-X`	HTTP method set karna (GET, POST, PUT, DELETE, PATCH)
`-H`	Custom header bhejna
`-d`	Request body/data bhejna
`-i`	Response headers + body dono dikhana
`-I`	Sirf headers dikhana (HEAD request)
`-v`	Verbose — pura connection detail dikhana
`-s`	Silent — progress bar hata dena
`-o`	Output ek file mein save karna
`-O`	URL ke naam se hi file save karna
`-L`	Redirect ko follow karna
`-u`	Username:password bhejna (basic auth)

11. Practice Karne Ke Liye

Yeh saare commands terminal (Linux/Mac) ya Windows PowerShell/CMD (curl already built-in hai) mein try kar sakte ho:

- GET se saare todos list karo: curl https://jsonplaceholder.typicode.com/todos
- Sirf ek user ke todos: curl https://jsonplaceholder.typicode.com/todos?userId=1
- Headers dekho -i lagakar
- Apna khud ka POST body bana kar bhejo

Tip: Jab confuse ho jao ki kaunsa option use karna hai, terminal mein `curl --help` ya `man curl` chala kar dekh sakte ho.